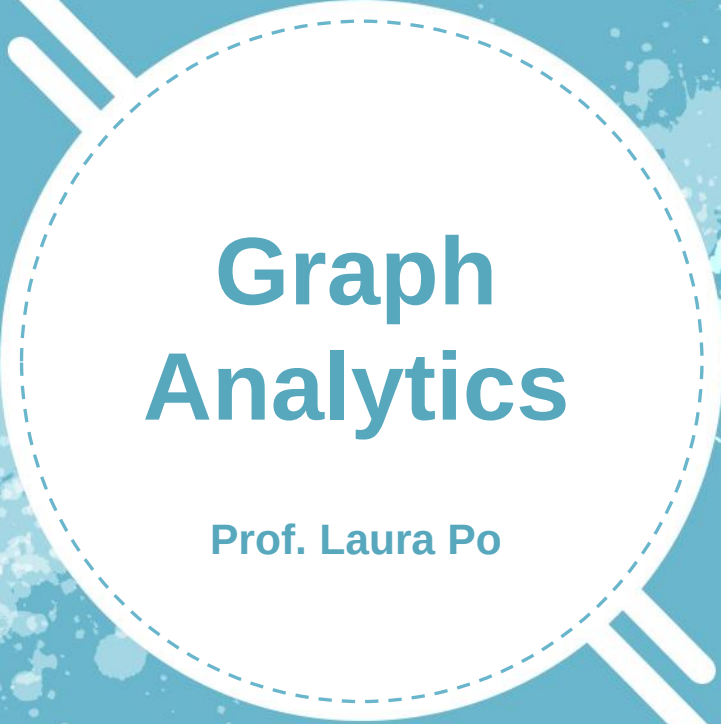




Graph Analytics

Prof. Laura Po

Slides are partially taken from the course "Web Data Science – Graph Relevance Analysis " thought by Matteo Lissandrini - Aalborg University

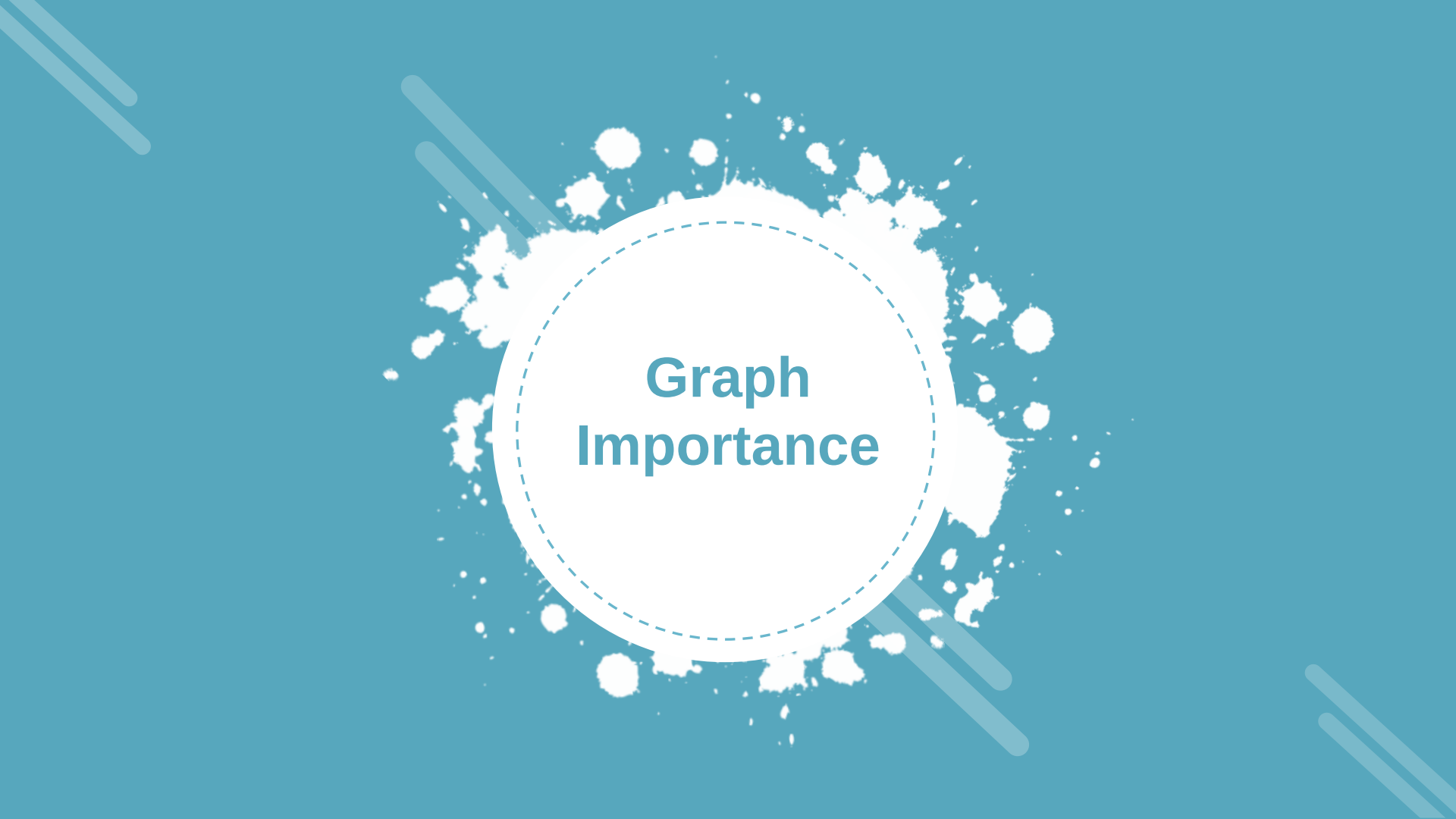
Recap

We want to «study» the graphs to obtain insights about:

- Structure & Connectivity: is it well connected? Are there clusters of nodes (communities)?
- Importance of single objects: which node/edge has important role (position) in the graph?

Importance:

- How many connections?
- How many nodes can reach this node?
- How many «paths» pass through this node?

The background is a solid teal color. It features several white, irregular splatters or paint-like marks scattered across the surface. There are also several thin, light blue diagonal lines running across the background, some parallel and some intersecting.

Graph Importance

Advanced methods to understand the importance of a node

Random Walk & Markov Model

1. Random Walk
2. Transition Probability
3. Teleport
4. Markov Model
5. Algebraic representation

Basic

Concepts:

Some core concepts to understand first

Relevance Measures

1. Co-Citation
2. HITS
3. Page Rank
4. Personalized Page Rank

Applications:

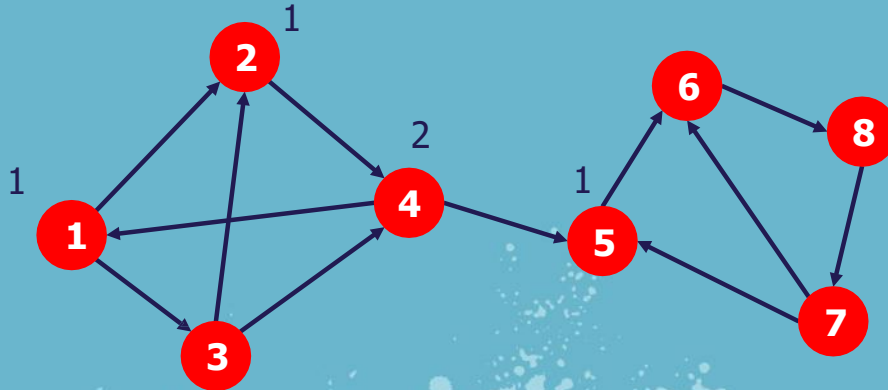
How to use the *basic concepts* to obtain the answers we need

The Random walk

1. Pick a node
2. Select a neighbour at random: take a step
3. Keep making steps until we are “tired”
4. Take note of the node where we stop and how often we visit each node



Random Walk: *traversal of the graph by selecting neighbours at random. It is possible to visit the same edge/node multiple times. We keep note of the “frequency” with which each node is visited*



Directed Graph:
We need to follow the directions of edges

The Random Walk (2)

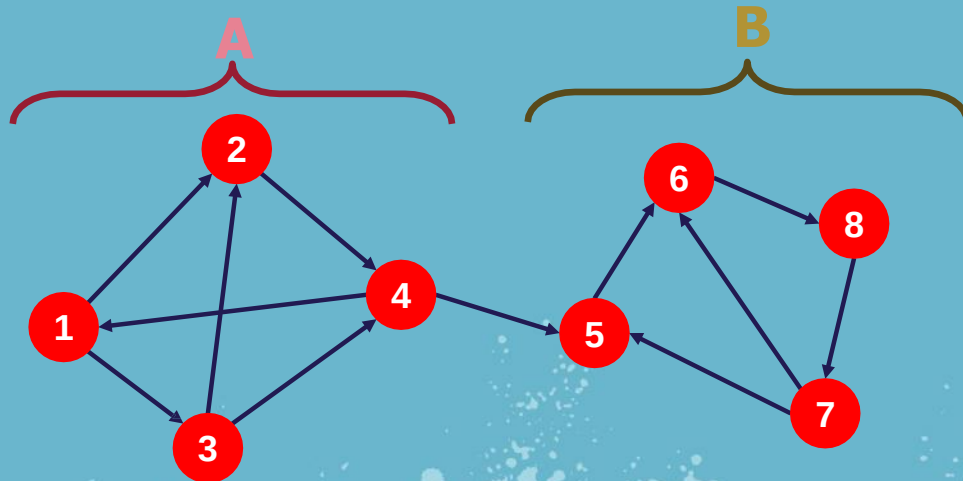
Let's play a game

1. I pick a random node (not telling which one)



2. I perform a random walk (not telling how many steps, let's say >3)

3. Your guess: where am I on the graph? Group A or Group B



The Random Walk (3)

Let's play a game

1. **I pick a random node** (not telling which one)



2. **I perform a random walk** (not telling how many steps, let's say >3)

3. Your guess: where am I on the graph? Group A or Group B

Probability

6 0.233843

8 0.222960

7 0.213162

5 0.139355

4 0.068738

2 0.046465

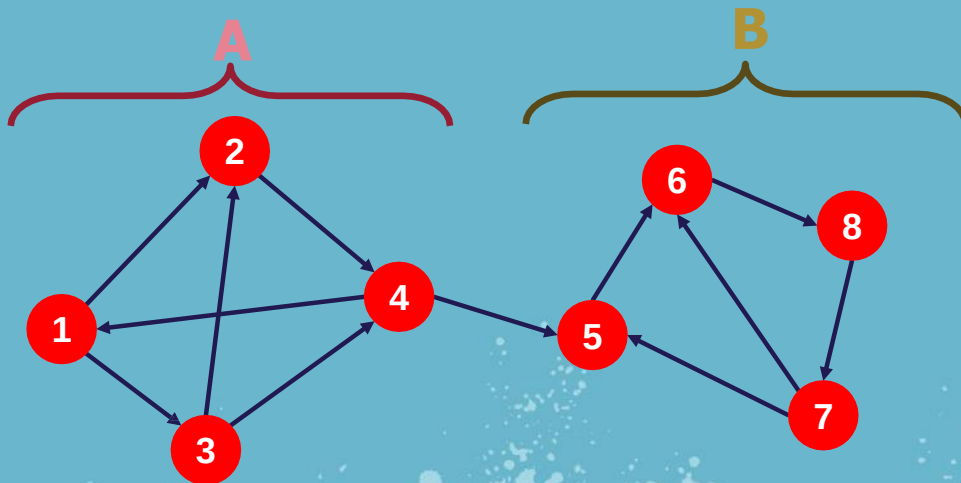
1 0.043432

3 0.032045

GROUP B ~ 0.81

*We will see
later how to
compute
these!*

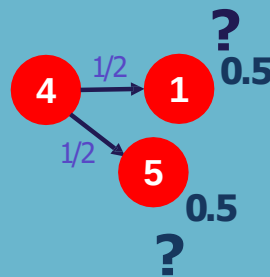
GROUP A ~ 0.19



Transition Probability

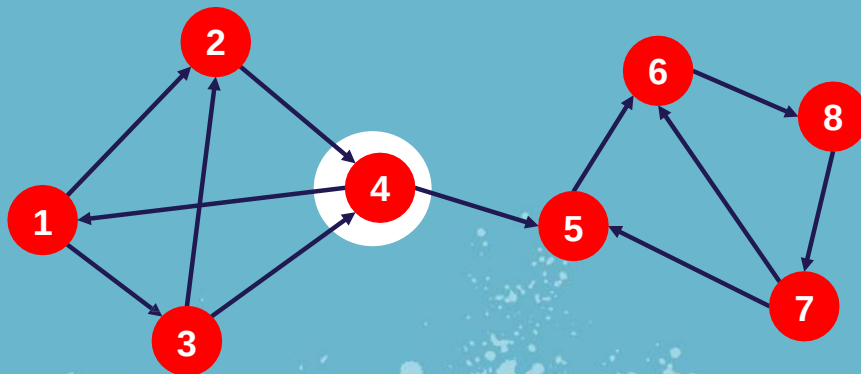
Given a node N_i , assuming to pick a random step, what is the probability to end up in its neighbour N_j ?

1. **If all edges are treated equally:** the probability depends on the number of outgoing edges, i.e., the degree of N_i – uniform transition probability
2. **Transition probabilities for all the edges of a single node always sum to 1**
3. **With weighted edges,** we can have non-uniform probabilities



Transition Probability

The probability, given a node N_i , to traverse its outgoing edge $e_{i,j}$ and reach neighbour node N_j

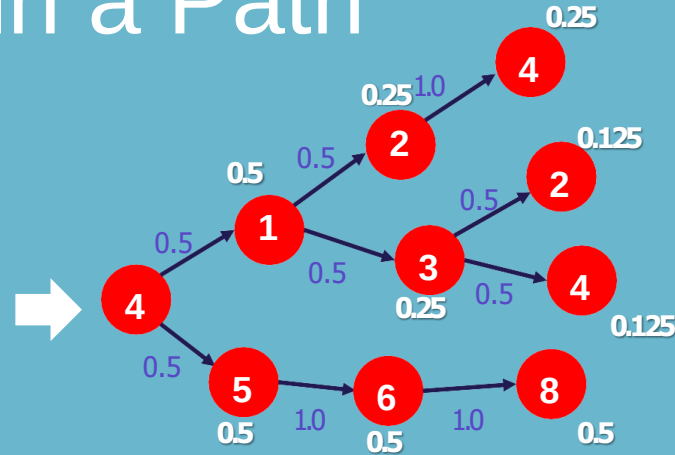


Transition Probability in a Path

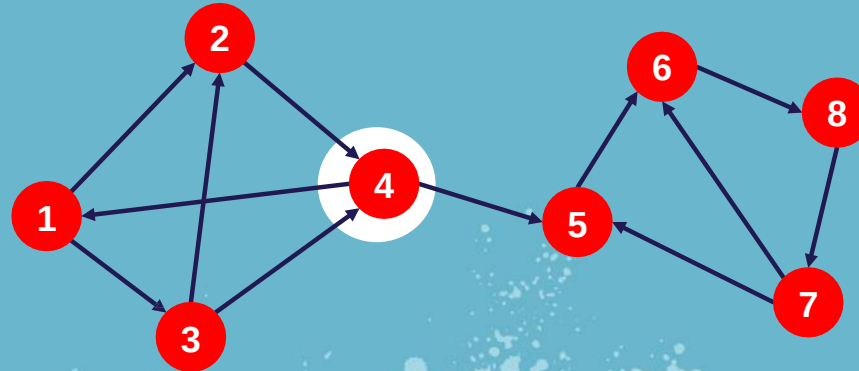
Given a node N_i , assuming to pick a series of K random steps, what is the probability to end up in a specific node N_j ?

We compute the joint probability over a path

What if we "get tired" before taking K steps?



Given $K=3$



Teleport Probability in a Random Walk

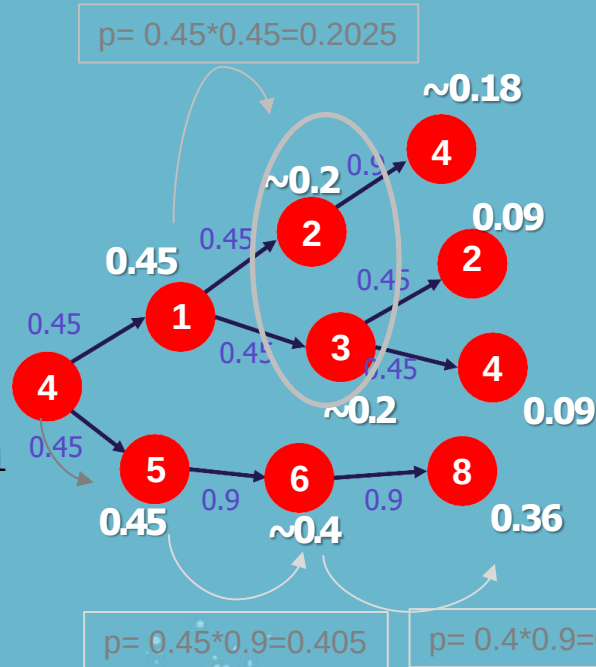
What if we "get tired" before taking K steps?

During a random walk, we model the probability of interrupting the walk with a parameter α

Transition probability + teleport probability = 1

Teleport Probability α : the probability of interrupting the random walk and of "jumping" to any other node at random.

Given Teleport $\alpha = 0,1$



The Random surfer

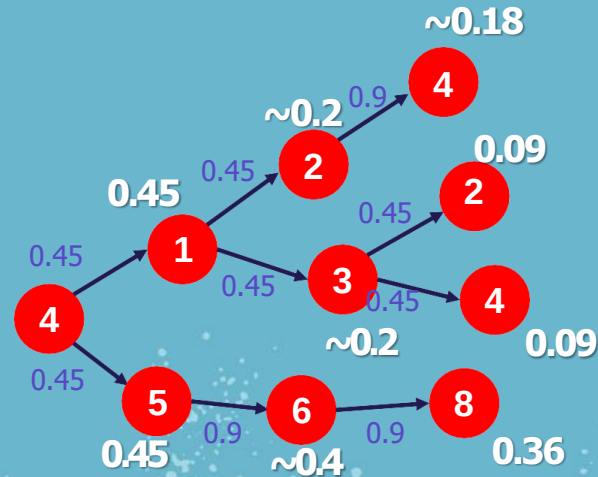
A model for a real use-case

The Random Walk model can represent the behaviour of users surfing the web

Random Surfer : *a web-user that starts from a page, follows links at random, after some clicks decides to start from a random new page*

Teleport Probability α : *the probability of interrupting the random walk and of “jumping” to any other node at random.*

Given Teleport $\alpha = 0,1$



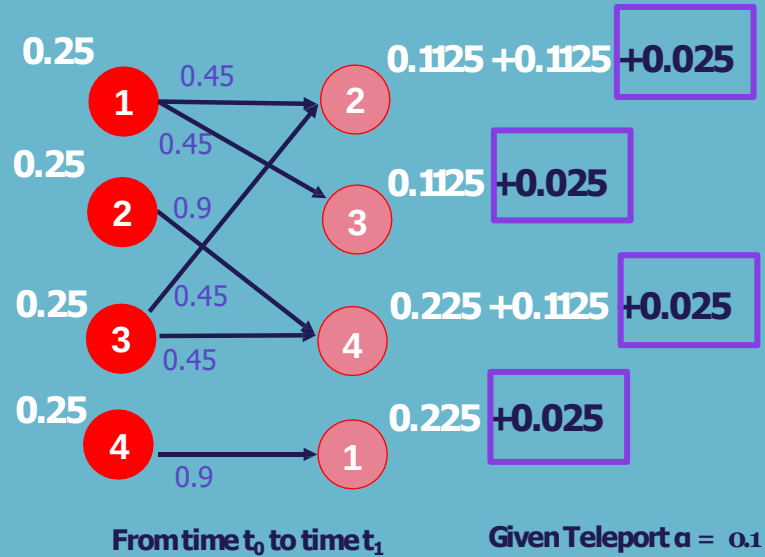
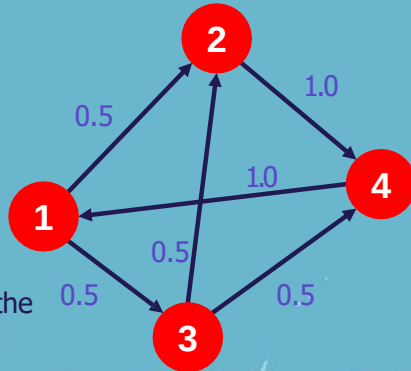
Markov Model

How do we model the random walk for each node

- Each node is a starting point with equal probability
- Each node is connected to the neighbours
- We set a teleport probability to end in any other node
- Edges have uniform transition probabilities
- We sum the transition probabilities of each node and the teleport probability (uniformly distributed)

Simpler Graph:

Using a simpler graph for now. Each edge presents the transition probability without accounting for the teleport probability



In this model the teleport probability can be seen also as “special ghost link”

Teleport Probability α : the probability of interrupting the random walk and of “jumping” to any other node at random.

Markov Model

*How do we model the random walk for each node:
We repeat the process iteratively*

The Markov Property: The future is independent of the past, given the present.

Markov chain: a stochastic process.

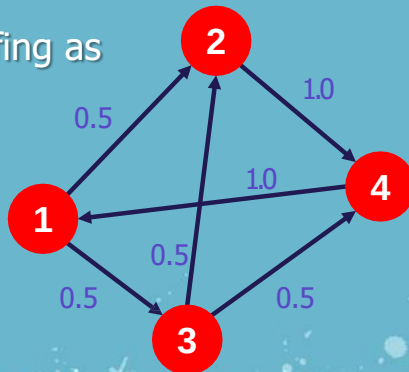
In the Markov chain, each node in the graph is regarded as a state.

An edge is a transition, which leads from one state to another state with a certain probability.

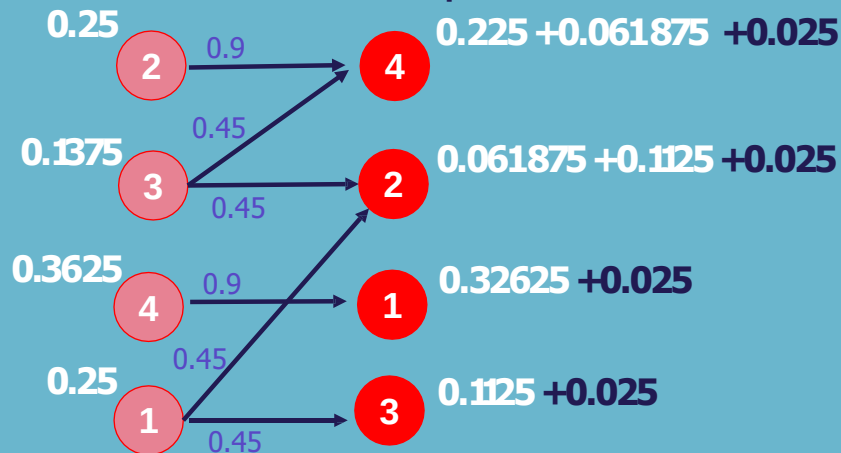
Markov chain models web surfing as a stochastic process

Simpler Graph:

Using a simpler graph for now.
Each edge presents the transition probability without accounting for the teleport probability



Given Teleport $\alpha = 0.1$



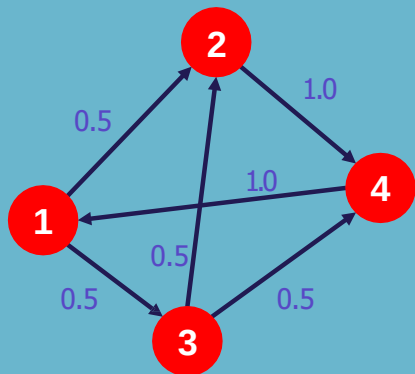
From time t_1 to time t_2

Intuition: Continue the iteration an infinite amount of time, what are the values? I.e., find the “**stationary distribution**”

Adjacency matrix / Transition Probability Matrix

Adjacency Matrix

- $A = N \times N$ matrix
- $A_{ij} = 1$ if edge between node i and j



A

	1	2	3	4
1	0	1	1	0
2	0	0	0	1
3	0	1	0	1
4	1	0	0	0

T

	1	2	3	4
1	0	$\frac{1}{2}$	$\frac{1}{2}$	0
2	0	0	0	1
3	0	$\frac{1}{2}$	0	$\frac{1}{2}$
4	1	0	0	0

Transition Probability Matrix

- $T = N \times N$ matrix
- $T_{ij} = P_{ij}$ transition probability from i to j
- **Rows always sum to 1**

The transition probability matrix is derived from A by normalizing the rows (row-normalized)

Algebraic Representation of the Markov model

Given the **Transition probability matrix** T of the graph, assuming no teleport probability ($\alpha = 0$)

Given the initial **vector of probabilities** v of each node

- **1 step of the process from time t_i to time t_{i+1} corresponds to the multiplication: $T^T \times v_i$**

Where T^T is the transpose matrix and v is column-normalized (sum of column = 1)

	1	2	3	4
1	0	$\frac{1}{2}$	$\frac{1}{2}$	0
2	0	0	0	1
3	0	$\frac{1}{2}$	0	$\frac{1}{2}$
4	1	0	0	0

T

	p
1	$\frac{1}{4}$
2	$\frac{1}{4}$
3	$\frac{1}{4}$
4	$\frac{1}{4}$

v_0



	1	2	3	4
1	0	0	0	1
2	$\frac{1}{2}$	0	$\frac{1}{2}$	0
3	$\frac{1}{2}$	0	0	0
4	0	1	$\frac{1}{2}$	0

T^T

$\times$

	p
1	$\frac{1}{4}$
2	$\frac{1}{4}$
3	$\frac{1}{4}$
4	$\frac{1}{4}$

v_0

$=$

	p
1	
2	
3	
4	

v_1

Algebraic Representation of the Markov model

Given the **Transition probability matrix** T of the graph, assuming no teleport probability ($\alpha = 0$)

Given the initial **vector of probabilities** v of each node

- **1 step of the process from time t_i to time t_{i+1} corresponds to the multiplication: $T^T \times v_i$**

Where T^T is the transpose matrix and v is column-normalized (sum of column = 1)

	1	2	3	4
1	0	$\frac{1}{2}$	$\frac{1}{2}$	0
2	0	0	0	1
3	0	$\frac{1}{2}$	0	$\frac{1}{2}$
4	1	0	0	0

T

	p
1	$\frac{1}{4}$
2	$\frac{1}{4}$
3	$\frac{1}{4}$
4	$\frac{1}{4}$

v_0

→

	1	2	3	4
1	0	0	0	1
2	$\frac{1}{2}$	0	$\frac{1}{2}$	0
3	$\frac{1}{2}$	0	0	0
4	0	1	$\frac{1}{2}$	0

T^T

×

	p
1	$\frac{1}{4}$
2	$\frac{1}{4}$
3	$\frac{1}{4}$
4	$\frac{1}{4}$

v_0

=

	p
1	$\frac{1}{4}$
2	$\frac{1}{4}$
3	$\frac{1}{8}$
4	$\frac{1}{4} + \frac{1}{8}$

v_1

Transition Matrix T : from incoming to outgoing (row normalized)

Transpose Matrix T^T : from outgoing to incoming

(column normalized) *it is also called the stochastic matrix but only if all columns sum to 1!*

Algebraic Representation of the Markov model (2)

Given the **Transition probability matrix** T & the initial **vector of probabilities** v of each node

We can account for **the teleport probability** α so that

- **1 step of the process from time t_i to time t_{i+1} corresponds to the multiplication:** $(1-\alpha)T^T \times v_i + \alpha \times v_o$

$$\alpha = 0.1$$

$$(1-\alpha) \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1/2 & 0 & 1/2 & 0 \\ 1/2 & 0 & 0 & 0 \\ 0 & 1 & 1/2 & 0 \end{bmatrix} \end{matrix} \times \begin{matrix} & p \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 1/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{bmatrix} \end{matrix} + \alpha \begin{matrix} & p \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 1/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{bmatrix} \end{matrix} = \begin{matrix} & p \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 1/4 \\ 1/4 \\ 29/80 \\ 11/80 \end{bmatrix} \end{matrix}$$

$T^T \qquad v_o \qquad v_o \qquad v_1$

During iteration,
the vector for the tele
port stays the same.
We update only the
vector multiplying the
matrix

Algebraic Representation of the Markov model (3)

Given the **Transition probability matrix** T & the initial **vector of probabilities** $\mathbf{v}$ of each node

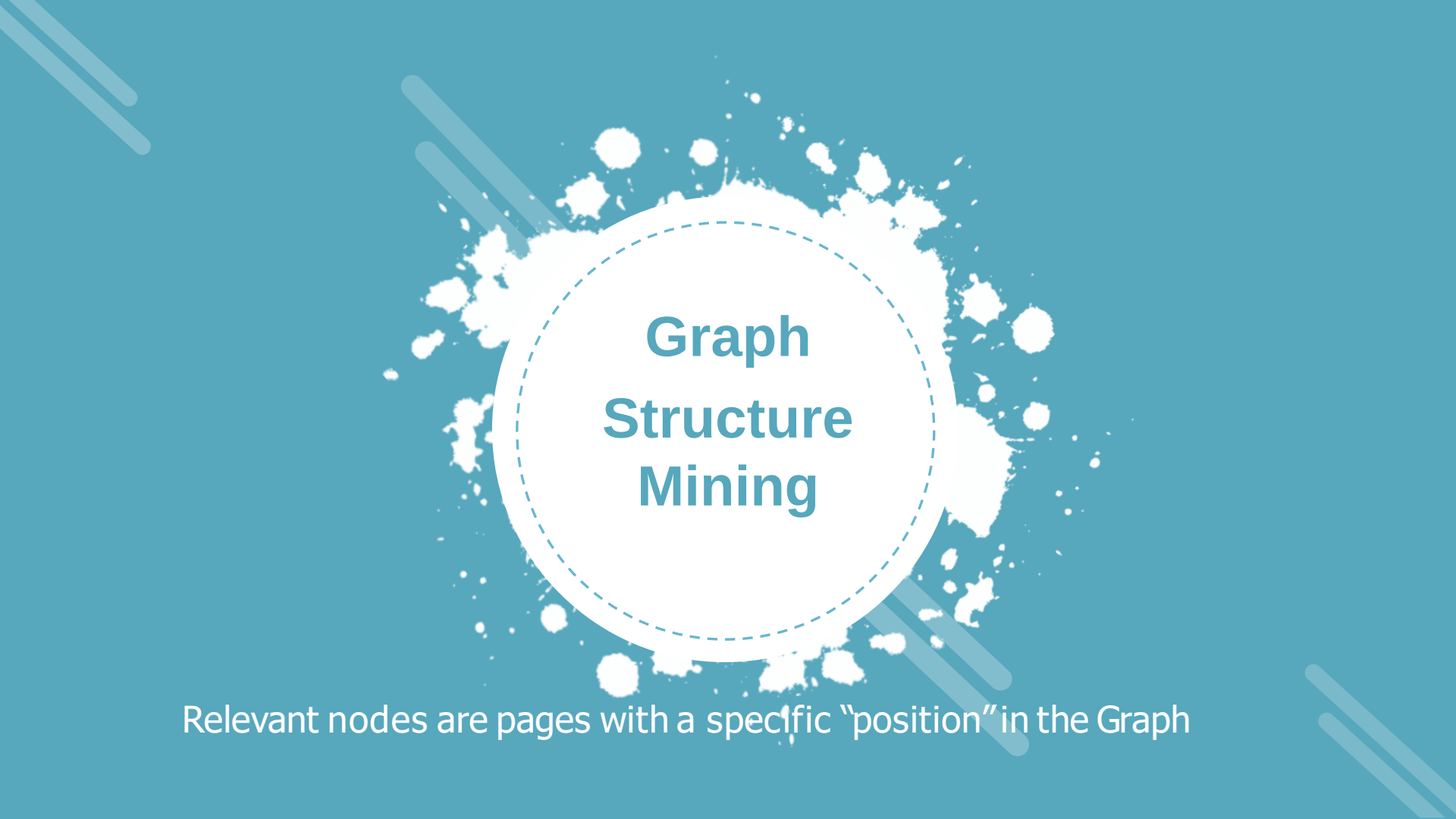
We can account for **the teleport probability** α so that

- **1 step of the process from time t_i to time t_{i+1} corresponds to the multiplication:**

$$(1 - \alpha) \cdot \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \dots & a_{nn} \end{bmatrix}^T \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}_{t_i} + \alpha \cdot \begin{bmatrix} \frac{1}{n} \\ \vdots \\ \frac{1}{n} \end{bmatrix} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}_{t_{i+1}}$$

We refer to this process as Power Iteration:

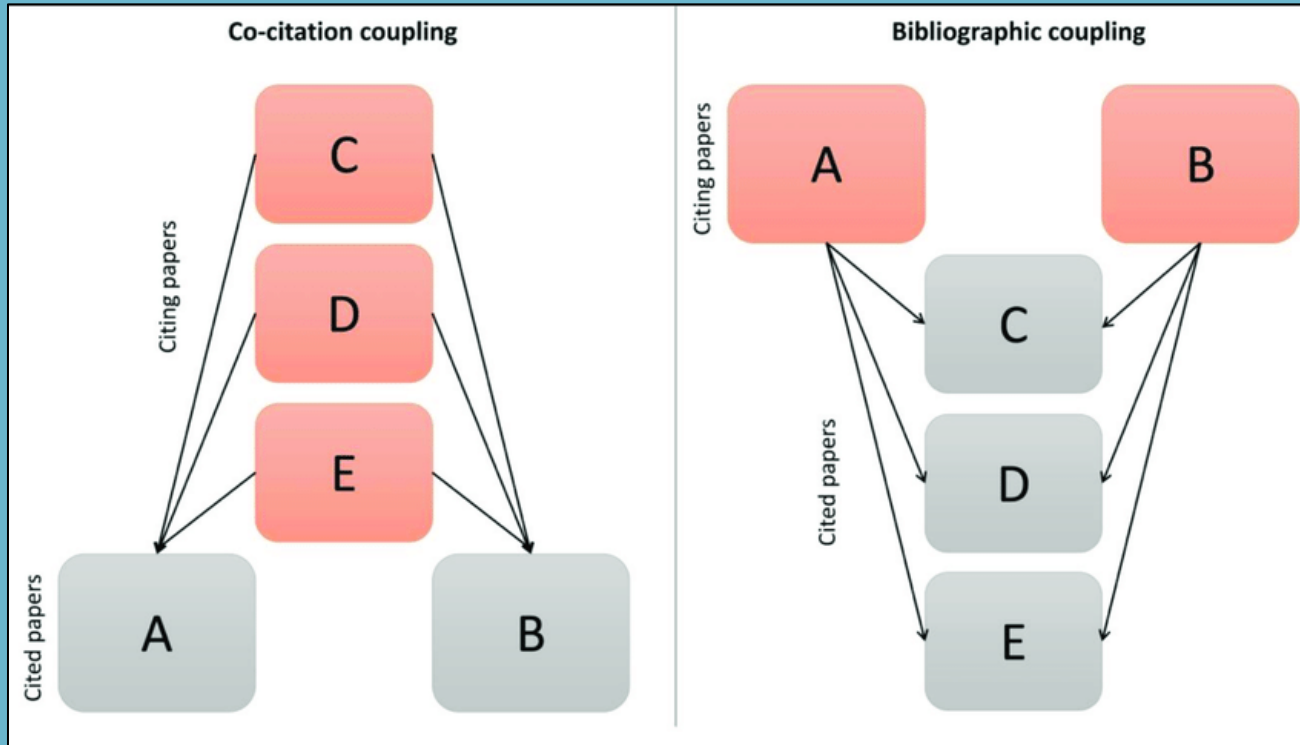
When we compute the vector at time t_{i+1} using the multiplication with the vector at time t_i



Graph Structure Mining

Relevant nodes are pages with a specific “position” in the Graph

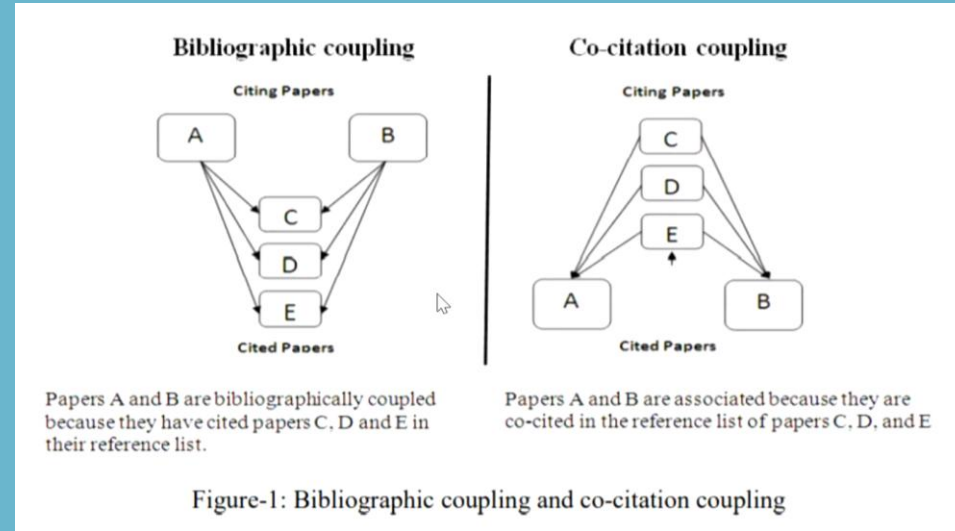
Co-Citation and Bibliographic Coupling



Co-Citation and Bibliographic Coupling

Citation Network: a directed graph representing scientific publications as nodes, an edge goes from A to B if A has a reference to B (A citation!).

- This is a *directed acyclic graph* (DAG)
- When a paper cites another paper, a relationship is established between the publications
- *Citation analysis uses these relationships (links) to perform various types of analysis.*
- *We discuss two types of citation analysis: co-citation and bibliographic coupling.*



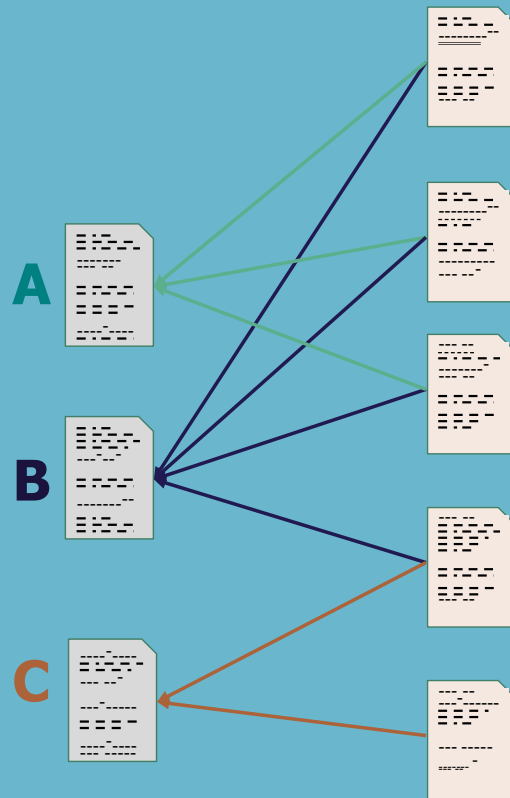
Co-Citation

Citation Network: a directed graph representing scientific publications as nodes, an edge goes from A to B if A has a reference to B (A citation!).

- If papers **i** and **j** are both cited by paper **k**, then **they may be related** in some sense to one another.
- → Given two papers **i** and **j**: **The more papers** they are cited by, **the stronger** their relationship is.

Co-citation

similarity: Which pair of papers is more likely to be about a similar set of topics? (A,B) or (B,C)?



Their references ← Papers

Co-Citation Matrix

- Let $\mathbf{L}$ be the citation matrix. Each cell of the matrix is defined as :

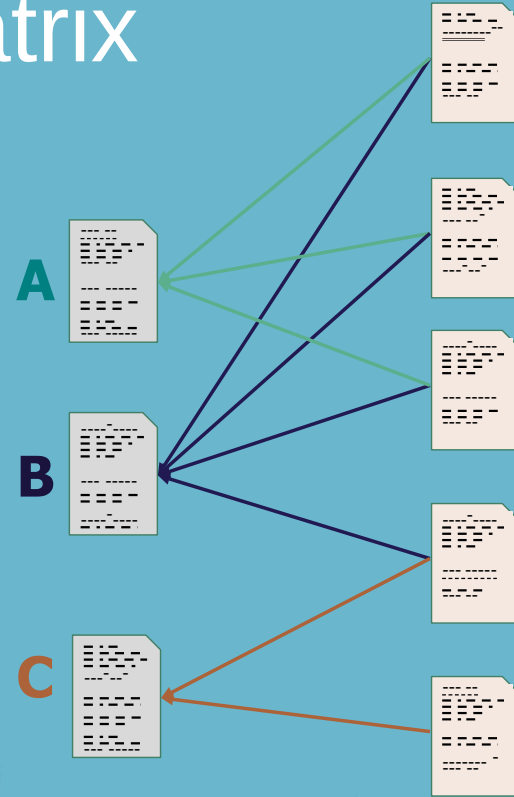
$L_{ij} = 1$ if paper i cites paper j , and 0 otherwise.

- Co-citation** (denoted by C_{ij})

is a similarity measure defined as **the number of papers that co-cite i and j** ,

$$C_{ij} = \sum_{k=1}^n L_{ki} L_{kj},$$

- C_{ij} is the number of papers that cite i
- A square matrix $\mathbf{C}$ can be formed with C_{ij} and it is called the **co-citation matrix**.



Co-Citation Computation

Compute the Adjacency Matrix of the graph

$L_{ij} = 1$ if paper i cites paper j , and 0 otherwise – $L_{BC}=1$.

	A	B	C	D	E	F
A	0	0	0	0	0	0
B	0	0	1	0	0	0
C	0	0	0	0	0	0
D	1	1	0	0	1	0
E	1	1	0	0	0	1
F	0	1	1	0	0	0

L

	A	B	C	D	E	F
A	0	0	0	1	1	0
B	0	0	0	1	1	1
C	0	1	0	0	0	1
D	0	0	0	0	0	0
E	0	0	0	1	0	0
F	0	0	0	0	1	0

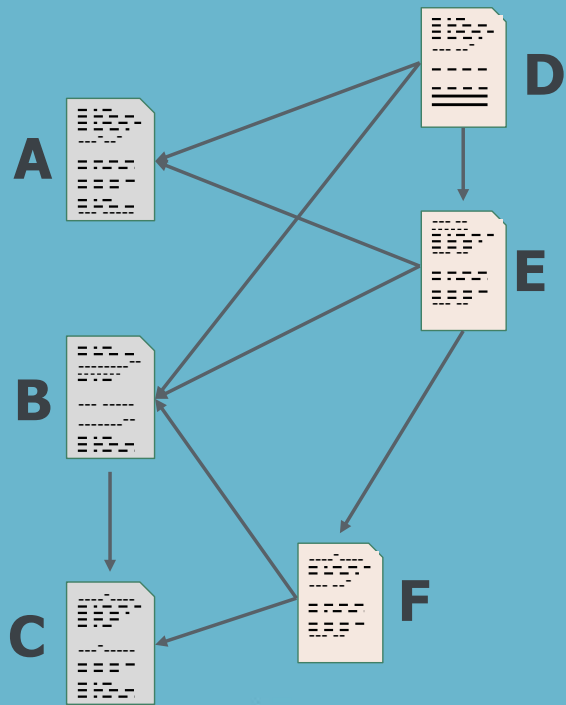
L^T

	A	B	C	D	E	F
A	2	2	0	0	1	1
B	2	3	1	0	1	1
C	0	1	2	0	0	0
D	0	0	0	0	0	0
E	1	1	0	0	1	0
F	1	1	0	0	0	1

C

To obtain the full matrix, one can compute **$C = L^T \times L$**

$$C_{ij} = \sum_{k=1}^n L_{ki} L_{kj},$$



Co-citation computation:
How many papers cite B and C?

Bibliographic Coupling

Bibliographic coupling occurs when two works reference a common third work in their bibliographies. It is an indication that a probability exists that the two works treat a related subject matter.

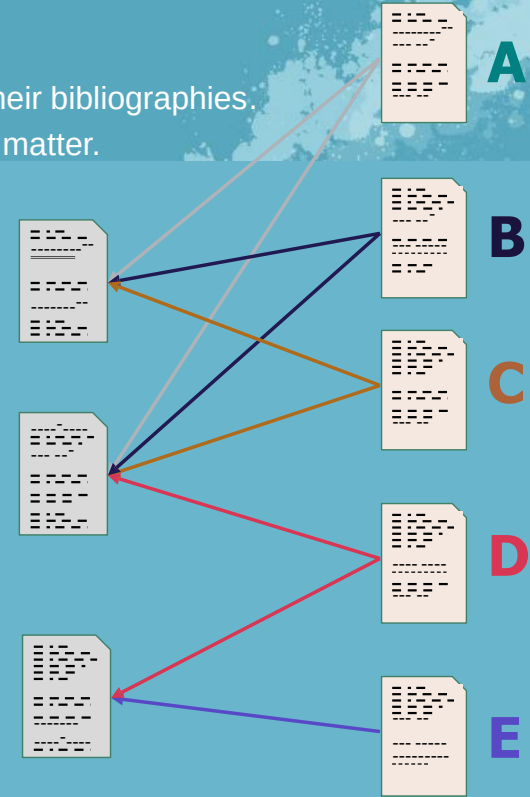
- Bibliographic coupling operates on a similar principle to Co-citation
- Bibliographic coupling links papers **that cite the same articles**
if papers i and j **both cite paper k** , they may be related
- The more papers they both cite, the stronger their similarity is
- We use B_{ij} to represent the number of papers that are cited by both papers i and j

$$B_{ij} = \sum_{k=1}^n L_{ik} L_{jk}$$

- B_{ii} is the number of references (in the reference list) of paper i .
- A square matrix $\mathbf{B}$ can be formed with B_{ij}
- it is called **the bibliographic coupling (co-reference) matrix**, to obtain it the process is similar to what done for the co-citation matrix

Additional material:

[https://www.leydesdorff.net/aca/#:~:text=Co%2Dcitation%20or%20co%2Dauthor,two%20authors\)%20appear%20to%20be.](https://www.leydesdorff.net/aca/#:~:text=Co%2Dcitation%20or%20co%2Dauthor,two%20authors)%20appear%20to%20be.)



Bibliographic coupling:

Which pair of papers is more likely to be about a similar set of topics? (A,B) or (A,D)?

Bibliographic Coupling Computation

B_{ij} to represent the number of papers that are cited by both papers i and j .

	A	B	C	D	E	F
A	0	0	0	0	0	0
B	0	0	1	0	0	0
C	0	0	0	0	0	0
D	1	1	0	0	1	0
E	1	1	0	0	0	1
F	0	1	1	0	0	0

L

	A	B	C	D	E	F
A	0	0	0	1	1	0
B	0	0	0	1	1	1
C	0	1	0	0	0	1
D	0	0	0	0	0	0
E	0	0	0	1	0	0
F	0	0	0	0	1	0

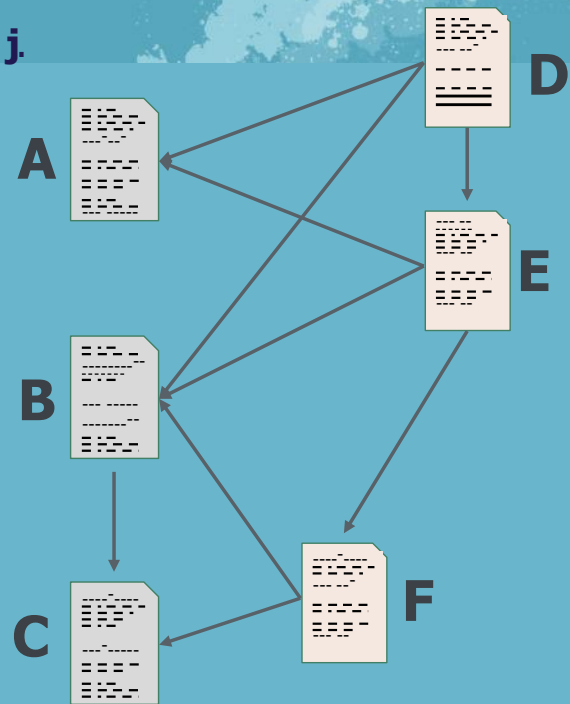
L^T

	A	B	C	D	E	F
A	0	0	0	0	0	0
B	0	1	0	0	0	1
C	0	0	0	0	0	0
D	0	0	0	3	2	1
E	0	0	0	2	3	1
F	0	1	0	1	1	2

B

To obtain the full matrix, one can compute $B = L \times L^T$

$$B_{ij} = \sum_{k=1}^n L_{ik} L_{jk}$$



HITS: Hypertext Induced Topic Search

HITS (Hypertext Induced Topic Search) is a link analysis algorithm developed by Jon Kleinberg in 1999 as an alternative approach to PageRank for ranking web pages.

The algorithm introduces two important concepts:

1. Authority and Hub Scores

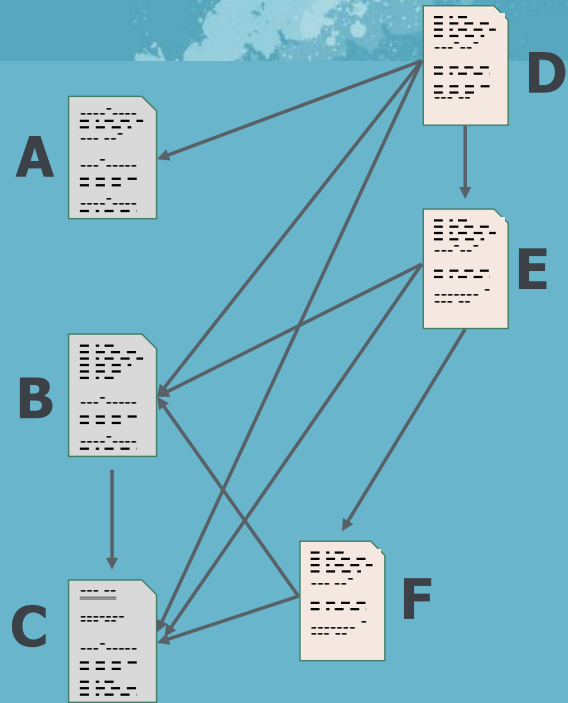
HITS assigns two scores to each page:

- ❑ **Authority Score:** Measures the value of the content of the page itself
- ❑ **Hub Score:** Measures the value of the page's links to other pages

2. Mutual Reinforcement Relationship

The core insight of HITS is the mutual reinforcement between hubs and authorities:

- ❑ Good authorities are pages that are linked to by many good hubs
- ❑ Good hubs are pages that link to many good authorities

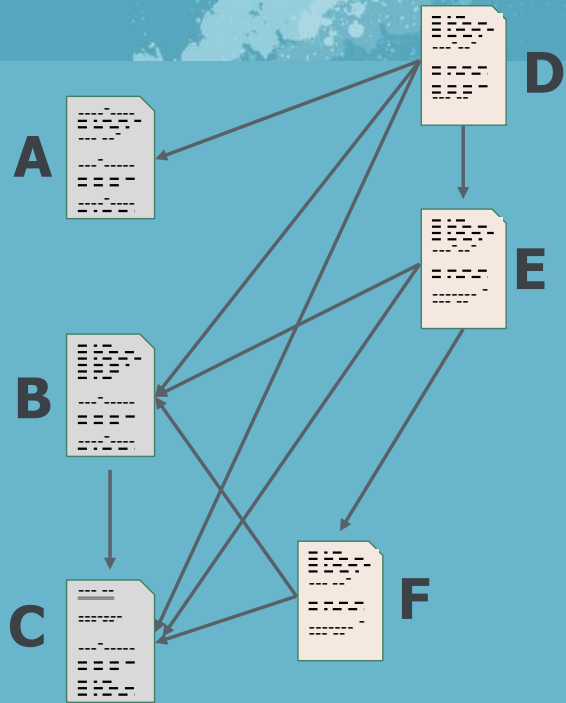


HITS: Hypertext Induced Topic Search

HITS is **search query dependent**

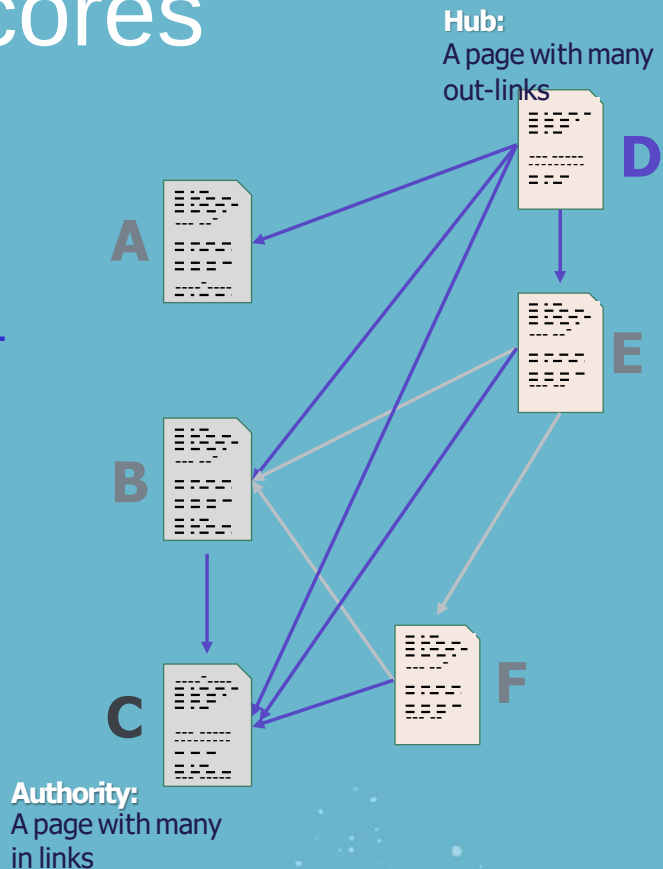
When the user issues a search query:

1. **HITS first expands the list of relevant pages** returned by a search engine (it retrieves all the pages linked to and linked by the result pages)
 2. **then produces two rankings** of the expanded set of pages, authority ranking and hub ranking.
- **Authority:** an authoritative page is a page *with many in-links*.
 - The idea is that the page may have good or authoritative content on some topic and thus many people trust it and link to it.
 - **Hub:** A hub is a page with *many out-links*.
 - The page serves as an organizer of the information on a particular topic (an index) and points to many good authority pages on the topic.



Hub/Authority Scores

- A **good hub** points to many **good authorities**
- A **good authority** is pointed to by many good hubs.
- Authorities and hubs have a **mutual reinforcement relationship**
- We model the graph as bipartite between hubs and authorities (a **bipartite sub-graph**).
- Nodes have both an authority and hub score → **Each node is repeated twice**

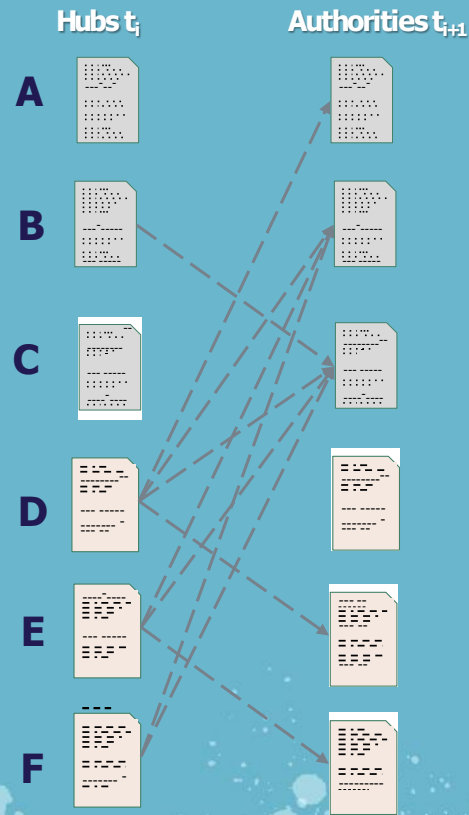


Hub/Authority Scores

We proceed iteratively

1. First, at time t_0 , we assign an authority score and a hub score to all nodes
(initial score $\frac{1}{\sqrt{N}}$)
2. Then (from time t_0 to time t_1) we accumulate the score from each authority to each hub and from each hub to each authority

We repeat the process from each time t_i to time t_{i+1} updating both values at each step



Hub/Authority Score computation

- Each page i has 2 scores:

- Authority score: a_i
- Hub score: h_i

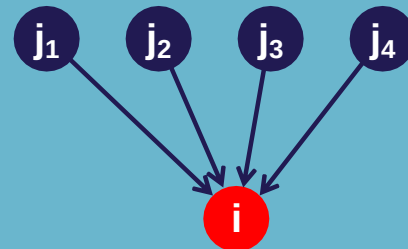
- Initialize:** $a_j^{(0)} = 1/\sqrt{N}$, $h_j^{(0)} = 1/\sqrt{N}$

- Then keep iterating until **convergence**:

- $\forall i$: Authority: $a_i^{(t+1)} = \sum_{j \rightarrow i} h_j^{(t)}$
- $\forall i$: Hub: $h_i^{(t+1)} = \sum_{i \rightarrow j} a_j^{(t)}$
- $\forall i$: Normalize:

We want that at the end it holds

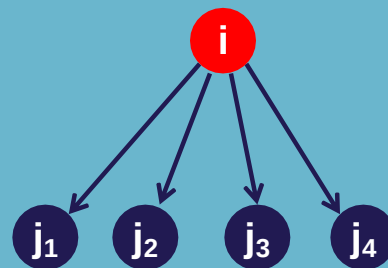
$$\sum_i \left(a_i^{(t+1)}\right)^2 = 1, \sum_j \left(h_j^{(t+1)}\right)^2 = 1$$



$$a_i = \sum_{j \rightarrow i} h_j$$

Do you see the connection
with Co-citation and
Bibliographic coupling?

We can use the co-citation
and co-reference matrix



$$h_i = \sum_{i \rightarrow j} a_j$$

Hub/Authority Score computation (2)

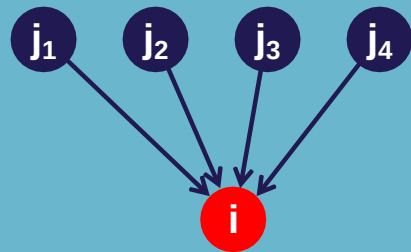
HITS converges to a single stable point!

- **Notation:**

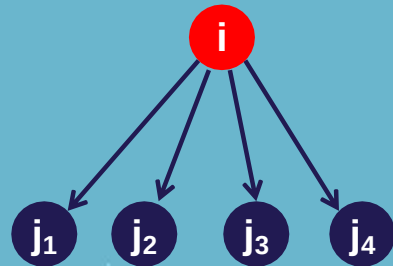
- N web pages
- Vectors $\mathbf{a} = (a_1 \dots, a_n)$ - authority score, $\mathbf{h} = (h_1 \dots, h_n)$ hub score
- Adjacency matrix (called Link matrix of the Web)
- $A (N \times N)$: $A_{ij} = 1$ if there is a link $i \rightarrow j$, 0 is not otherwise
- Then $\mathbf{h}_i = \sum_{j \rightarrow i} \mathbf{a}_j$ can be rewritten as $\mathbf{h}_i = \sum_j A_{ij} \mathbf{a}_j$ $\mathbf{h} = A \cdot \mathbf{a}$
- Similarly, $\mathbf{a}_i = \sum_{j \rightarrow i} \mathbf{h}_j$ can be rewritten as $\mathbf{a}_i = \sum_j A_{ji} \cdot \mathbf{h}_j$ $\mathbf{a} = A^T \cdot \mathbf{h}$
- **Repeat until convergence:**
 - $\mathbf{h} = A \cdot \mathbf{a}$ $\mathbf{a} = A^T \cdot \mathbf{h}$
 - Normalize $\mathbf{a}$ and $\mathbf{h}$

This approach is called
"power iteration"

- Then: $\mathbf{a} = A^T \cdot \underbrace{(A \cdot \mathbf{a})}_{\text{new } \mathbf{h}}$
 $\underbrace{\hspace{1.5cm}}_{\text{new } \mathbf{a}}$



$$a_i = \sum_{j \rightarrow i} h_j$$

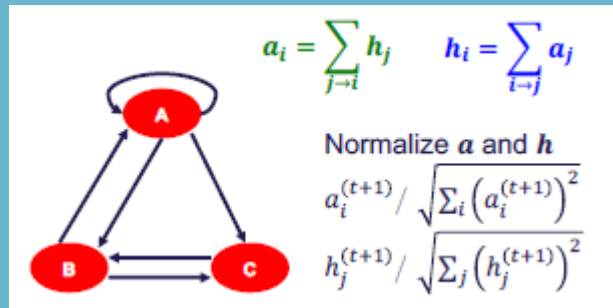


$$h_i = \sum_{i \rightarrow j} a_j$$

Hub/Authority Score computation (3)

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\mathbf{A}^T = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$



$h(A)$	=	.58	.80	.80	.79	...	.788
$h(B)$	=	.58	.53	.53	.57	...	.577
$h(C)$	=	.58	.27	.27	.23	...	.211
$a(A)$	=	.58	.58	.62	.62	...	.628
$a(B)$	=	.58	.58	.49	.49	...	.459
$a(C)$	=	.58	.58	.62	.62	...	.628

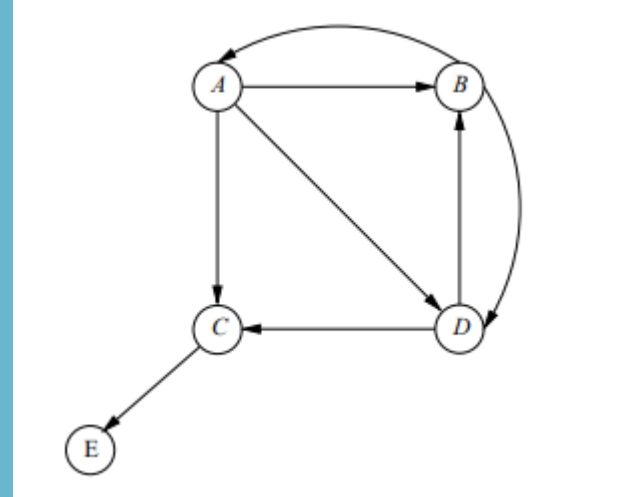
Convergence Guarantee:

Under reasonable assumptions about $\mathbf{A}$,
HITS converges to vectors $\mathbf{h}^$ and $\mathbf{a}^*$*

HITS follows the Markov-model
can you see why?

Exercise

- Compute hub and authority score of the graph



Solution

$$L = \begin{bmatrix} 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad L^T = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix}$$

$$\mathbf{h} = \begin{bmatrix} 1 \\ 0.3583 \\ 0 \\ 0.7165 \\ 0 \end{bmatrix} \quad \mathbf{a} = \begin{bmatrix} 0.2087 \\ 1 \\ 1 \\ 0.7913 \\ 0 \end{bmatrix}$$

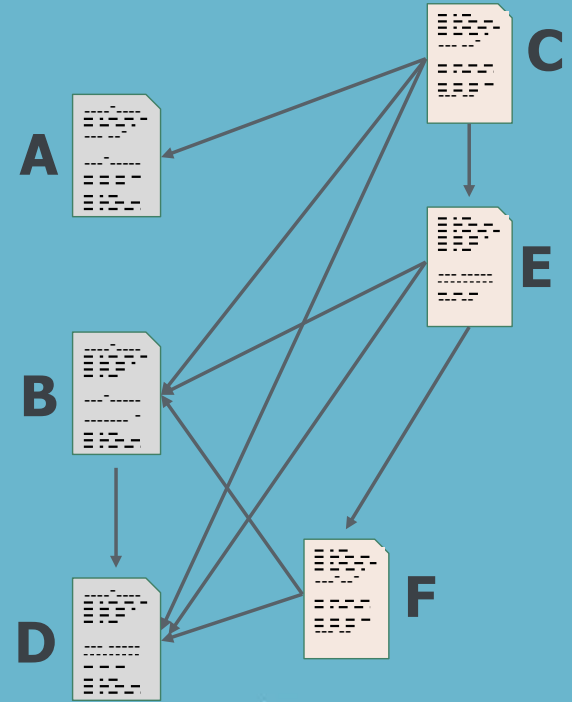
Page Rank

PageRank is an algorithm that measures the importance of nodes (pages) in a graph by analyzing the structure of links between them. Originally developed by Google founders Larry Page and Sergey Brin in 1996-1997 at Stanford University when they were PhD students, it revolutionized web search by determining which web pages are most valuable.

Basic concept: A page is important if it receives links from other important pages. This creates a recursive definition where a page's importance depends on the importance of pages linking to it.

Mathematical model: PageRank represents the web as a directed graph where:

- Websites are nodes
- Links between sites are edges
- The "importance" of each page is distributed through these links



Page Rank

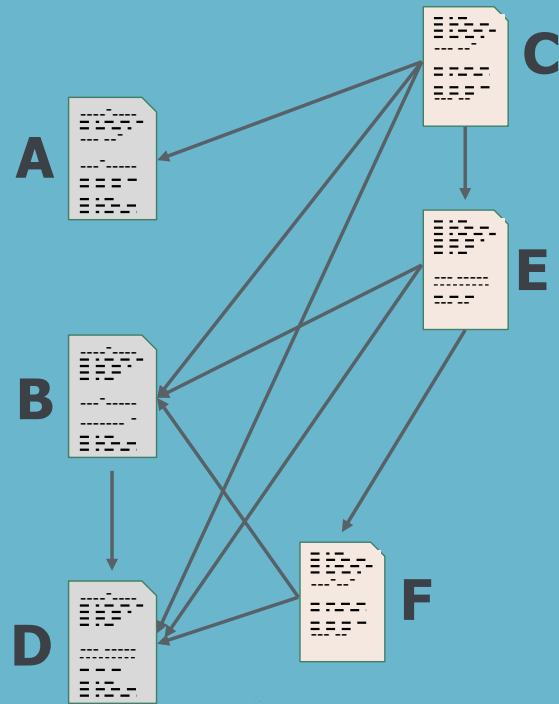
Intuition:

- links from page i to page j is a **vote** from i to j
- Votes from pages with **a lot of votes count more**

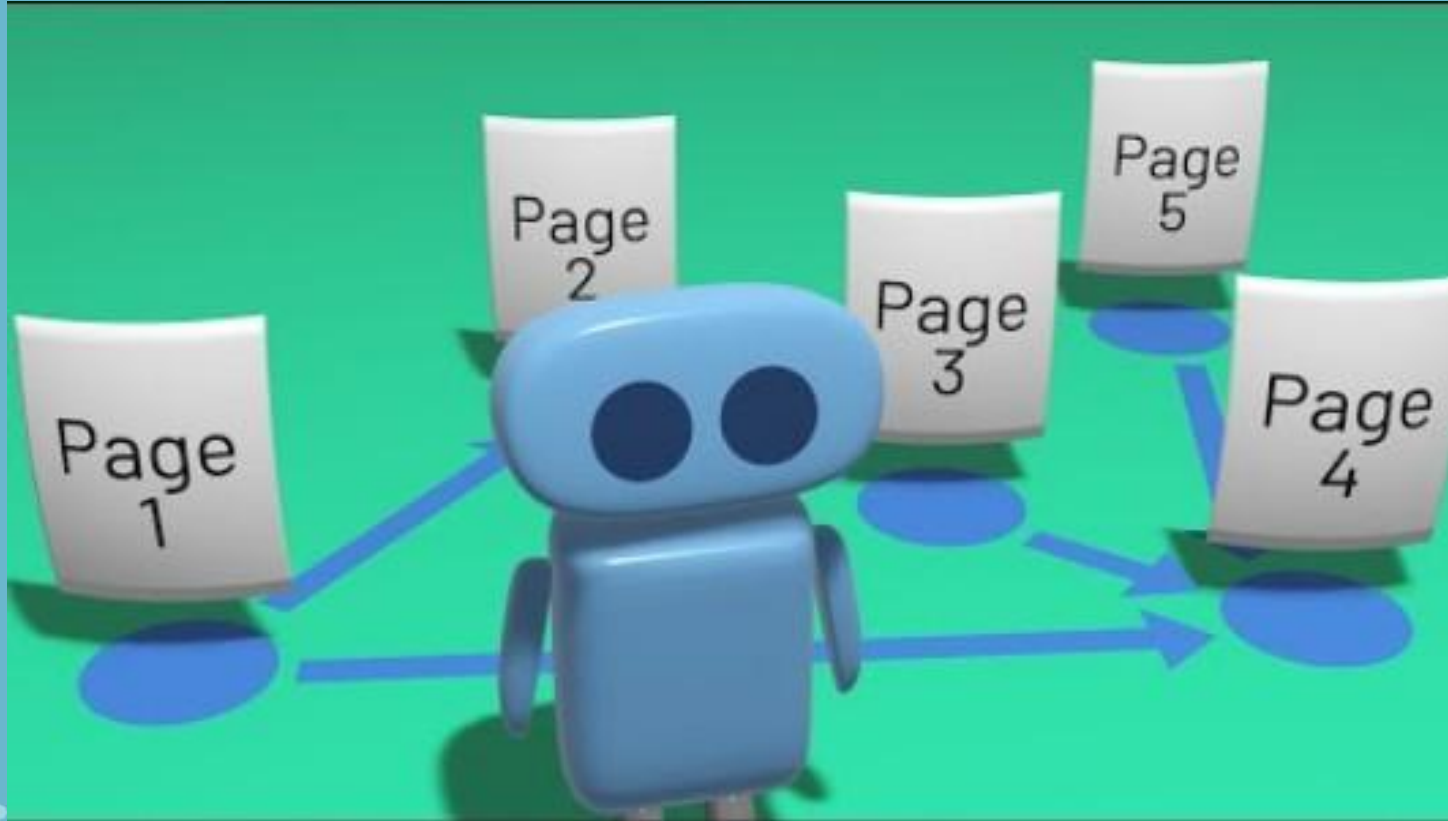
In other words

1. Each link's vote is proportional to the **importance** of its source page
2. If page j with importance r_j has n out-links, each link gets r_j / n votes
3. Page j own importance is the sum of the votes on its **in-links**

$$\text{Rank}(B) = \text{Rank}(C)/4 + \text{Rank}(E)/3 + \text{Rank}(F)/2$$



How Google's PageRank Algorithm Works



<https://www.youtube.com/watch?v=meonLcN7LD4>

Page Rank: Importance Flow

Intuition:

- A “vote” from an important page is worth more
- A page is important if it is pointed to by other important pages

Define a “rank” r_j for page j

$$r_j = \sum_{i \rightarrow j} \frac{r_i}{d_i}$$

r_i rank of the node i with a link $i \rightarrow j$
 d_i out-degree of node i

Rank “Flow” equations:

$$r_C = r_C / 2 + r_A / 2$$

$$r_A = r_C / 2 + r_B$$

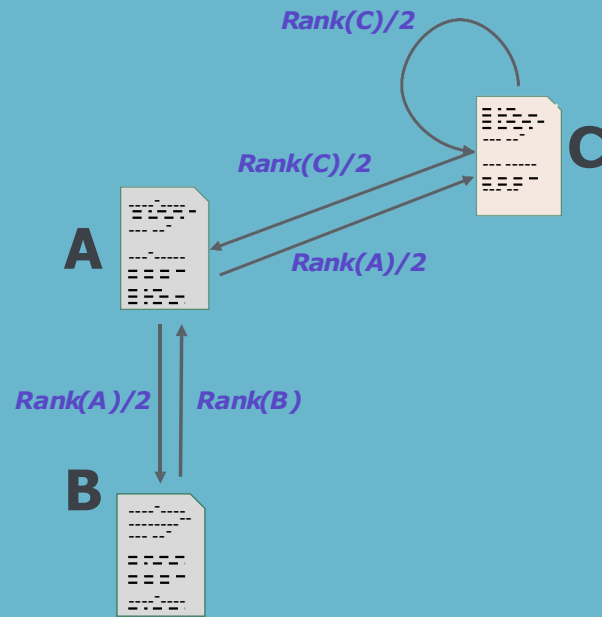
$$r_B = r_A / 2$$

To solve these equations, we can set

$$r_C + r_A + r_B = 1$$

and solve analytically to find $r_C = 2/5$ $r_A = 2/5$ $r_B = 1/5$

But we will see a better way!

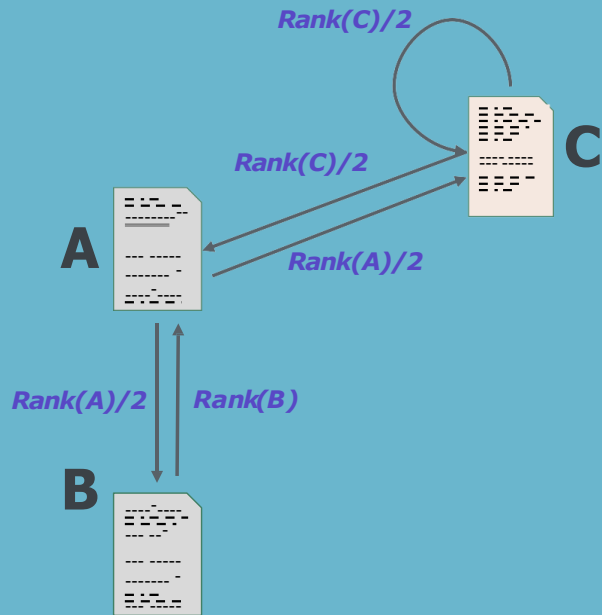


Page Rank: Matrix Formulation

- **Stochastic adjacency matrix M**
 - Let page i has d_i out-links
 - If $i \rightarrow j$, then $M_{ji} = \frac{1}{d_i}$ else $M_{ji} = 0$
 - M is a column stochastic matrix $\rightarrow$ Columns sum to 1
- **Rank vector r** : vector with an entry per page
 - r_i is the importance score of page i
 - and sums to $\sum_i r_i = 1$
- The flow equations can be written

$$r = M \cdot r$$

	A	B	C
A	0	1	$\frac{1}{2}$
B	$\frac{1}{2}$	0	0
C	$\frac{1}{2}$	0	$\frac{1}{2}$



$$r_j = \sum_{i \rightarrow j} \frac{r_i}{d_i}$$

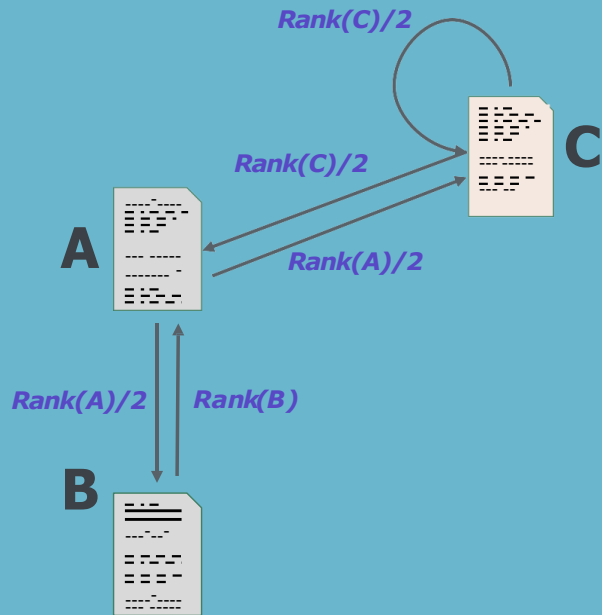
d_i ... out-degree of node i

Page Rank: Power Iteration

- **Given a graph with N nodes** where the nodes are pages and edges are hyperlinks
- **Power iteration:** a simple iterative scheme
 - Initialize the vector: $\mathbf{r}^{(0)} = [1/N, \dots, 1/N]^T$
 - Iterate: $\mathbf{r}^{(t+1)} = \mathbf{M} \cdot \mathbf{r}^{(t)}$
 - Stop when $|\mathbf{r}^{(t+1)} - \mathbf{r}^{(t)}|_1 < \varepsilon$

$$|\mathbf{x}|_1 = \sum_{r=1}^n |x_r|$$

L1 Norm: the sum of the magnitudes of the vectors in a space.



$$r_j = \sum_{i \rightarrow j} \frac{r_i}{d_i}$$

d_i ... out-degree of node i

Page Rank: Example

Power Iteration:

Set $r_i = 1/N$

1: $r'_j = \sum_{i \rightarrow j} \frac{r_i}{d_i}$ d_i out-degree of node i

2: $r = r'$

Go to **1**

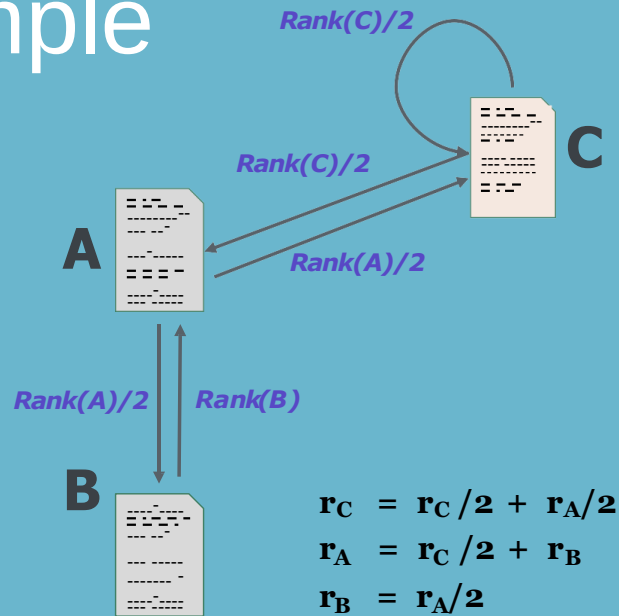
Example:

$$\begin{bmatrix} r_C \\ r_A \\ r_B \end{bmatrix} = \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix}$$

Iteration: 0, 1, 2, ...

Markov Model / Random Walk:

- (1) This computation follows the Markov model
- (2) This model is equivalent to a random walk with no teleport



Page Rank: Example

Power Iteration:

Set $r_i = 1/N$

1: $r'_j = \sum_{i \rightarrow j} \frac{r_i}{d_i}$ d_i out-degree of node i

2: $r = r'$

Go to **1**

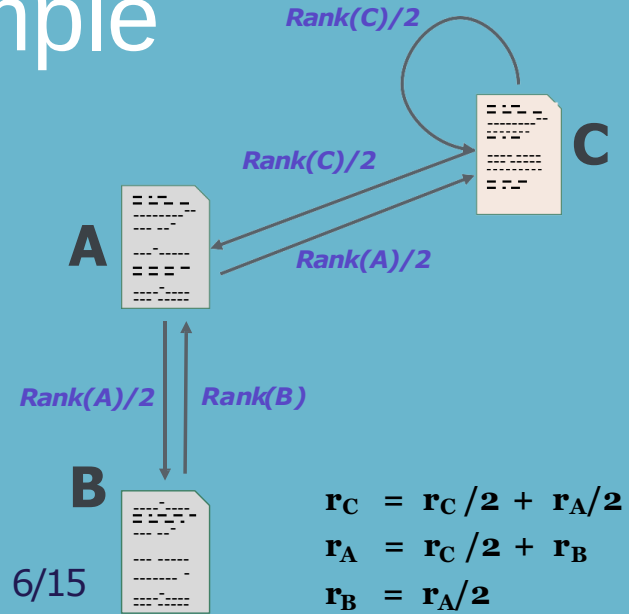
Example:

$$\begin{bmatrix} r_C \\ r_A \\ r_B \end{bmatrix} = \begin{bmatrix} 1/3 & 1/3 & 5/12 & 9/24 \\ 1/3 & 3/6 & 1/3 & 11/24 & \dots \\ 1/3 & 1/6 & 3/12 & 1/6 \end{bmatrix} \begin{bmatrix} 6/15 \\ 6/15 \\ 3/15 \end{bmatrix}$$

Iteration: 0, 1, 2, ...

Markov Model / Random Walk:

- (1) This computation follows the Markov model
- (2) This model is equivalent to a random walk with no teleport



$$r_C = r_C / 2 + r_A / 2$$

$$r_A = r_C / 2 + r_B$$

$$r_B = r_A / 2$$

The final result is the "stationary distribution", a probability distribution over pages, i.e., if we start in random node and traverse the graph randomly, what is the probability I am on node X?

Stationary Distribution

- Given a transition matrix (with normalized columns), following the power iteration method, are we guaranteed to stop the computation?

Existence and Uniqueness of Stationary Distribution in Markov Processes:

For graphs that satisfy certain conditions,

the stationary distribution is unique and eventually will be reached no matter what the initial probability distribution at time $t = 0$

Example of non-convergence:



$$\begin{matrix} r_a \\ r_b \end{matrix} = \begin{matrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{matrix}$$

Iteration 0, 1, 2, ...

Stationary Distribution: Convergence Issues

- Given a transition matrix (with normalized columns), following the power iteration method, are we guaranteed to obtain a meaningful result?

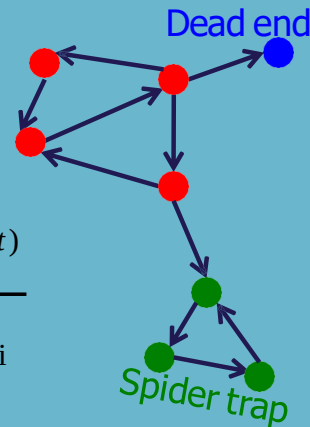
Example of problematic convergence:



$$\begin{matrix} r_a \\ r_b \end{matrix} = \begin{matrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{matrix}$$

Iteration 0, 1, 2, ...

$$r_j^{(t+1)} = \sum_{i \rightarrow j} \frac{r_i^{(t)}}{d_i}$$



Dead-end & spider-trap issues:

- 1. Dead-end:** a node with no out-links, a sink, the random walk reaching here will then stop and the "importance" will be lost in the next iteration
- 2. Spider trap:** nodes in a cycle with no exit, a loop, the random walk will iterate continuously without reaching a stable state

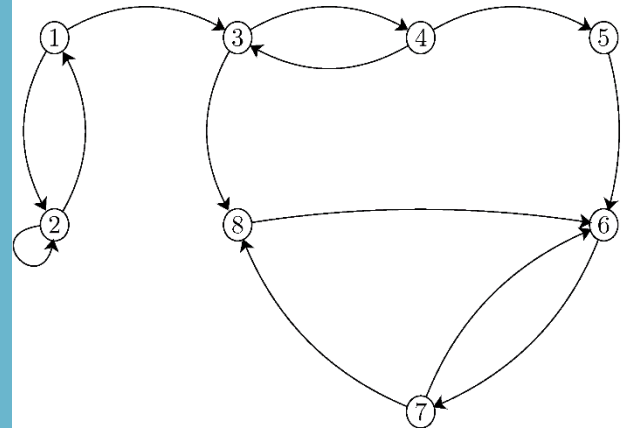
Page Rank: Appropriateness of the Markov Model

Existence and Uniqueness of Stationary Distribution in Markov Processes

The following conditions should hold:

1. *The transition matrix is a stochastic matrix: all columns sum to 1*
2. *The matrix is irreducible $\rightarrow$ the graph is strongly connected: $i \leftrightarrow j$ for each couple of nodes in the graph*
3. *The matrix is aperiodic $\rightarrow$ it is possible to come back to each node in a fixed number of steps*

How can we modify the graph to ensure these conditions?



Page Rank: Appropriateness of the Markov Model

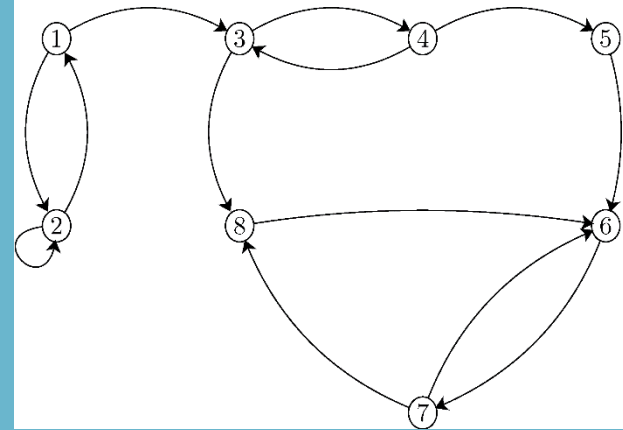
Existence and Uniqueness of Stationary Distribution in Markov Processes

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3. The matrix is aperiodic $\rightarrow$ it is possible to come back to each node in a fixed number of steps

Change the Transition Matrix to satisfy the Markov Model requirement: Add a link from each page to every page and give each link a small transition probability controlled by a parameter d .

← The teleport!



Page Rank: Appropriateness of the Markov Model

Change the Transition Matrix to satisfy the Markov Model requirement: Add a link from each page to every page and give each link a small transition probability controlled by a parameter d .

← *The teleport!*

- **Google's solution:**

At each step, random surfer has two options:

- With probability β , follow a link at random
- With probability $1-\beta$, jump to some random page

← β is called the **damping factor**
Usually between 0.8 and 0.95

- **PageRank equation** [Brin-Page, 98]

$$r_j = \sum_{i \rightarrow j} \beta \frac{r_i}{d_i} + (1 - \beta) \frac{1}{N}$$

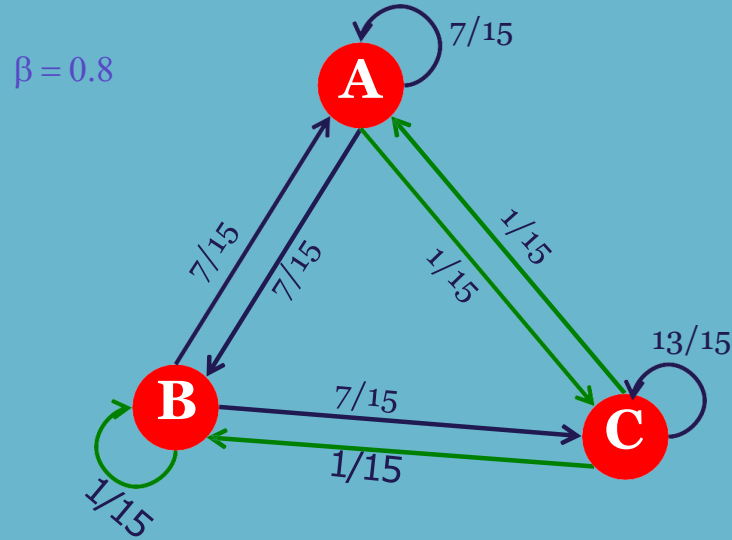
This formulation assumes that M has **no dead ends**. We can either preprocess matrix M to remove all dead ends or explicitly follow random teleport links with probability $1/N$ from dead-ends.
Easy solution: if a column has all 0 then replace with $1/N$

Dead end!



	A	B	C
A	0	0	$\frac{1}{2}$
B	$\frac{1}{2}$	0	0
C	$\frac{1}{2}$	0	$\frac{1}{2}$

Page Rank: Example Computation



$$0.8 \begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 0 & 0 \\ 0 & 1/2 & 1 \end{bmatrix} + 0.2 \begin{bmatrix} 1/3 & 1/3 & 1/3 \\ 1/3 & 1/3 & 1/3 \\ 1/3 & 1/3 & 1/3 \end{bmatrix}$$

Fully connected Matrix

A	7/15	7/15	1/15
B	7/15	1/15	1/15
C	1/15	7/15	13/15

A

A	=	1/3	0.33	0.24	0.26	7/33
B		1/3	0.20	0.20	0.18	5/33
C		1/3	0.46	0.52	0.56	21/33

Page Rank: Rearranging the Equation

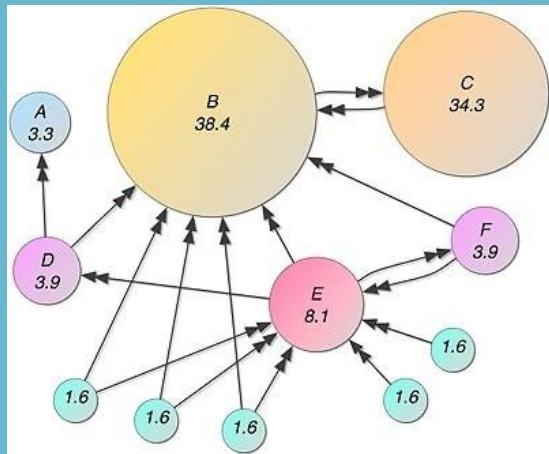
- $r = A \cdot r$, **where** $A_{ji} = Q M_{ji} + \frac{1-Q}{N}$
- $r_j = \sum_{i=1}^N A_{ji} \cdot r_i$
- $r_j = \sum_{i=1}^N \left[\beta M_{ji} + \frac{1-\beta}{N} \right] \cdot r_i$
- $= \sum_{i=1}^N \beta M_{ji} \cdot r_i + \frac{1-\beta}{N} \sum_{i=1}^N r_i$
- $= \sum_{i=1}^N \beta M_{ji} \cdot r_i + \frac{1-\beta}{N}$ **since** $\sum_{i=1}^N r_i = 1$
- **So we get:** $r = Q M \cdot r + \left[\frac{1-Q}{N} \right]_N$ $[x]_N \dots$ a vector of length N with all entries x

Note: Here we assume **M** has no dead-ends

Personalized Page Rank: Topic-Specific PageRank

- Page Rank measures a “generic” popularity of a page, is no specific for a search query or a topic
- Instead of generic popularity, can we measure popularity within a topic?
- **Goal:** Evaluate Web pages not just according to their popularity, but by how close they are to a particular topic, e.g., “sports” or “history”
- **Allows search queries to be answered based on interests of the user**
 - **Example:** A programmer looking for “*library for graph traversal*” wants different pages depending on the programming language they use the most

Personalized Page Rank: Topic-Specific PageRank



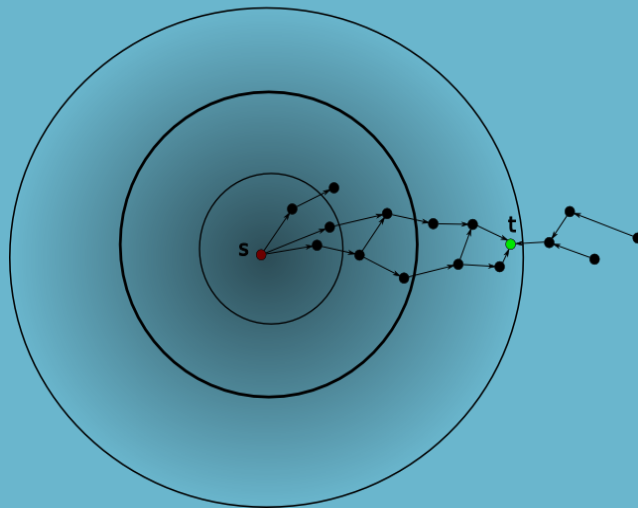
Global Page Rank

Starting from a random node, traversing randomly, **random restart point** anywhere in the graph

The role of the teleport: To avoid dead-end and spider-trap problems

Standard PageRank: Any page with equal probability

Topic Specific PageRank: A topic-specific set of "relevant" pages (**teleport set**)



Personalized Page Rank

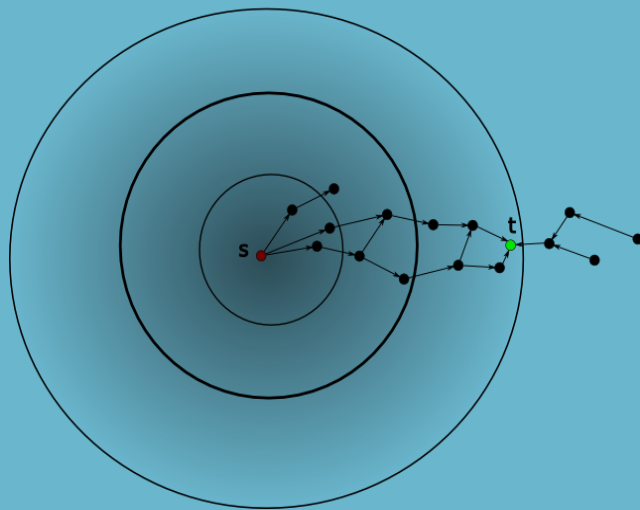
Starting from a **limited set of nodes**, traversing randomly, restart point is one in **the initial set**.
Bound not to travel too far

Personalized Page Rank: Topic-Specific PageRank

Idea: Bias the random walk

1. When walker teleports, she pick a page from a set S
2. The set S contains only pages that are **relevant to the topic**
E.g., pages with documentation of python libraries
3. For each teleport set S , we get a different vector r_s

When we teleport back to a single node is called: Random Walk with Restart



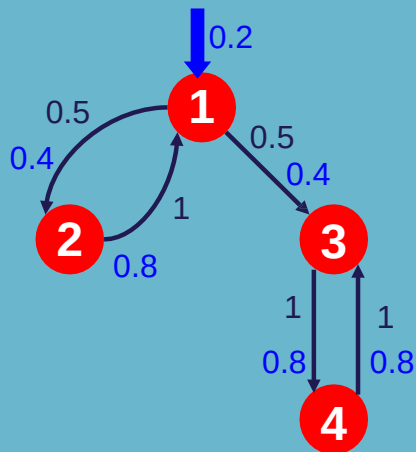
Personalized Page Rank

Starting from a **limited set of nodes**,
traversing randomly,
restart point is one in **the initial set**.
Bound not to travel too far

Personalized Page Rank: Example

$$A_{ij} = \begin{cases} Q M_{ij} + (1 - Q)/|S| & \text{if } i \in S \\ Q M_{ij} + 0 & \text{otherwise} \end{cases}$$

Suppose $S = \{1\}$
 $\beta = 0.8$



Node	Iteration				
	0	1	2	...	stable
1	0.25	0.4	0.28		0.294
2	0.25	0.1	0.16		0.118
3	0.25	0.3	0.32		0.327
4	0.25	0.2	0.24		0.261

$S = \{1\}$, $\beta = 0.90$:

$r = [0.17, 0.07, 0.40, 0.36]$

$S = \{1\}$, $\beta = 0.8$:

$r = [0.29, 0.11, 0.32, 0.26]$

$S = \{1\}$, $\beta = 0.70$:

$r = [0.39, 0.14, 0.27, 0.19]$

$S = \{1, 2, 3, 4\}$, $\beta = 0.8$:

$r = [0.13, 0.10, 0.39, 0.36]$

$S = \{1, 2, 3\}$, $\beta = 0.8$:

$r = [0.17, 0.13, 0.38, 0.30]$

$S = \{1, 2\}$, $\beta = 0.8$:

$r = [0.26, 0.20, 0.29, 0.23]$

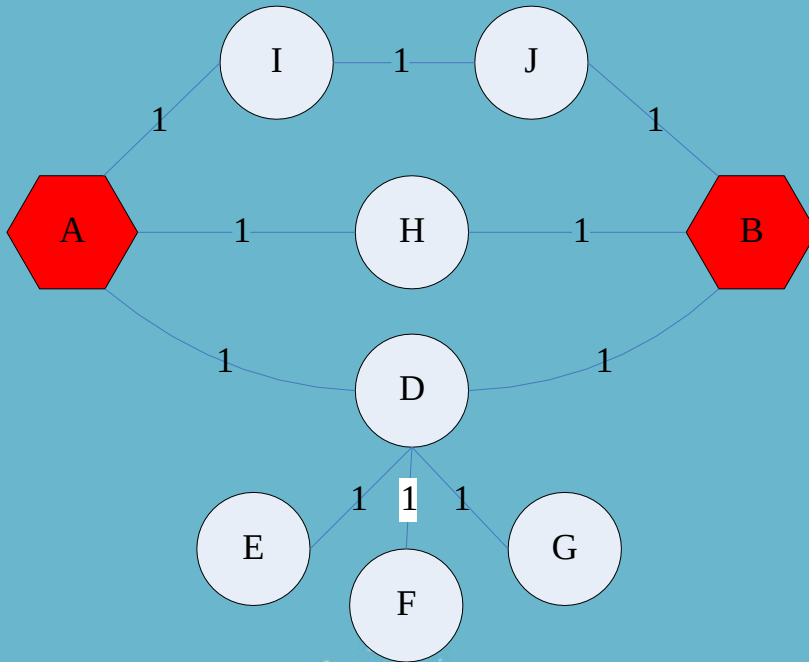
$S = \{1\}$, $\beta = 0.8$:

$r = [0.29, 0.11, 0.32, 0.26]$

Personalized Page Rank as a Proximity Measure

What is the probability to reach node B given that we start from node A?

What are the most "relevant" nodes for A ranked by "closeness"



- Multiple connections
- Quality of connection
 - Direct & Indirect connections
 - Length & "quantity"
- Degree: big hubs and the small world property!

a.k.a.: Relevance, Closeness, 'Relatedness'...

Example: Movie Recommendation

Imagine a **movie recommendation graph**, where:

- **Nodes** = Movies
- **Edges** = Links based on user similarity or genre overlap

If you want to recommend movies similar to "Inception," you could:

- Set the **personalization vector** to give weight to "Inception" and related movies.
- Run Personalized PageRank to rank the most relevant movies in the graph!



Thank you